

1. Let $P(n)$ be the proposition that 3 divides $5^{2n} - 1$ for all positive integers n .
 - BASIS STEP: $P(1)$ is true because $5^{2(1)} - 1 = 24$ which is divisible by 3
 - INDUCTIVE STEP: The inductive hypothesis assumes that $P(k)$ is true for any arbitrary integers $k \geq n$, where every integer k makes it so that $5^{2k} - 1$ is divisible by 3.
 - Using $P(k)$ to show that $P(k+1)$ is also true:

$$5^{2(k+1)} - 1 = 5^{2k+2} - 1 = 5^2 \cdot 5^{2k} - 1$$
 - Since $5^2 - 1 = 24$ which is divisible by 3, the rest of the term would also be divisible by 3 because the 5^{2k} term is being multiplied by a multiple of 3.
 - $\therefore 5^{2n} - 1$ is divisible by 3, for every positive integer n . ■

2. Let $P(n)$ be the proposition that $\sum_{j=0}^n 5j + 2 = \frac{5}{2}n^2 + \frac{9}{2}n + 2$, for all positive integers n .
 - BASIS STEP: $P(0)$ is true because $\sum_{j=0}^0 5j + 2 = \frac{5}{2}(0)^2 + \frac{9}{2}(0) + 2$ which is $2 = 2$
 - INDUCTIVE STEP: The inductive hypothesis assumes that $P(k)$ is true for any arbitrary integers $k \geq n$, where every integer k makes it so $\sum_{j=0}^k 5j + 2 = \frac{5}{2}k^2 + \frac{9}{2}k + 2$ holds true.
 - Using $P(k)$ to show that $P(k+1)$ is also true:

$$\begin{aligned} \sum_{j=0}^{k+1} 5j + 2 &= \frac{5}{2}(k+1)^2 + \frac{9}{2}(k+1) + 2 \\ \sum_{j=0}^{k+1} 5j + 2 &= \frac{5}{2}(k^2 + 2k + 1) + \frac{9}{2}(k+1) + 2 \\ \sum_{j=0}^{k+1} 5j + 2 &= \frac{5}{2}k^2 + 5k + \frac{5}{2} + \frac{9}{2}k + \frac{9}{2} + 2 \\ \sum_{j=0}^{k+1} 5j + 2 &= \frac{5}{2}k^2 + \frac{19}{2}k + 9 \end{aligned}$$

Now subtract the previous term...

$$\sum_{j=0}^{k+1} 5j + 2 - \sum_{j=0}^k 5j + 2 = \frac{5}{2}k^2 + \frac{19}{2}k + 9 - \frac{5}{2}k^2 + \frac{9}{2}k + 2$$

$$\therefore 5k + 7 = 5k + 7$$
 - $\therefore \sum_{j=0}^n 5j + 2 = \frac{5}{2}n^2 + \frac{9}{2}n + 2$, for all positive integers n . ■